

EE016-3-3-PP1 Project Phase 1 – Investigation in Engineering Project

Chapter 2 : Literature Review

Topic Learning Outcomes

At the end of this topic, you should be able to:

1. Investigate research problems and methods through literature review evidenced by a critical thinking framework such as 5W1H. (C4, PLO4)

Contents & Structure

- Literature Review

Recap From Last Lesson

- Introduction to the study
- Research Problem
- Aim
- Objectives
- Justification for this Research

Key Terms You Must Be Able To Use

If you have mastered this topic, you should be able to use the following terms correctly in your assessments:

- Literature Review
- Good sourcing of review
- Writing a review

CHAPTER 2 – LITERATURE REVIEW

Outline

2.1 Introduction

2.2 Literature Review

2.3 Summary

CHAPTER 2 – LITERATURE REVIEW

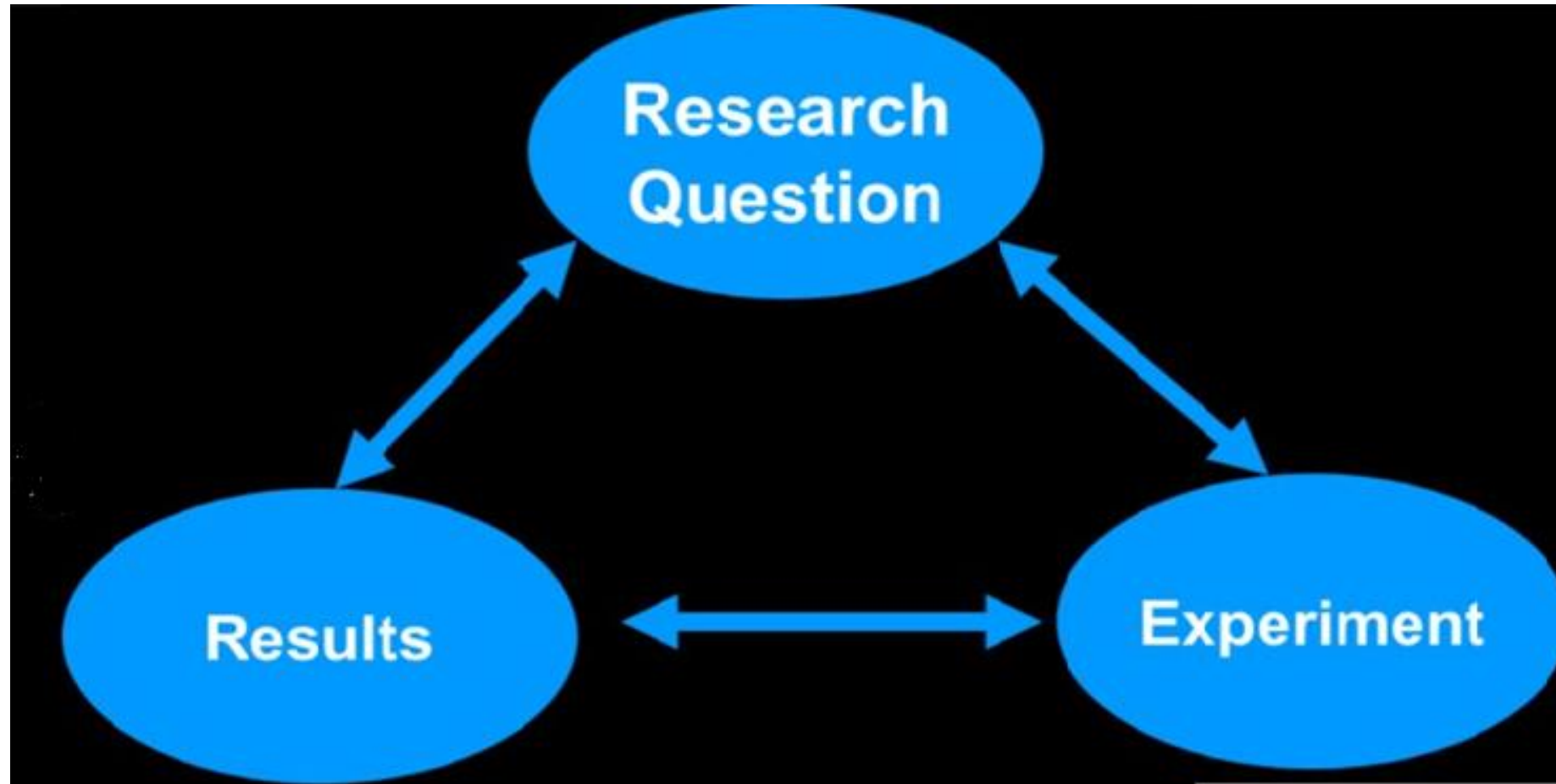
2.1 Introduction

- Give a brief background about the chapter.

CHAPTER 2 – LITERATURE REVIEW

“Researchers almost never conduct a study in an intellectual vacuum: their studies are undertaken within the context of an already existing knowledge base. Researchers generally undertake a literature review to familiarise themselves with that knowledge base”. (Polit & Hungler, 2000)

CHAPTER 2 – LITERATURE REVIEW



CHAPTER 2 – LITERATURE REVIEW

What is Literature Review?

- A literature review is a description of the literature. A literature review is a survey of everything that has been written about a particular topic, theory, or research or it may stand on its own. Much more than a simple list of sources, an effective literature review analyzes and synthesizes information about key themes or issues.

CHAPTER 2 – LITERATURE REVIEW

What is Literature Review?

- The literature review is the process of consolidating the various strands of past research into a single narrative describing the evolution of the research domain.

CHAPTER 2 – LITERATURE REVIEW

Why do we do Literature Review?

- A literature review gives an overview of the field of inquiry:
 - what has already been said on the topic,
 - who the key writers are,
 - what the prevailing theories and hypotheses are,
 - what questions are being asked, and
 - what methodologies and methods are appropriate and useful,
 - What each source contribute to the topic,
 - What is the relationship between the various contributions.
 - critical literature review shows how prevailing ideas fit into the thesis, and how the thesis agrees or differs from them.

CHAPTER 2 – LITERATURE REVIEW

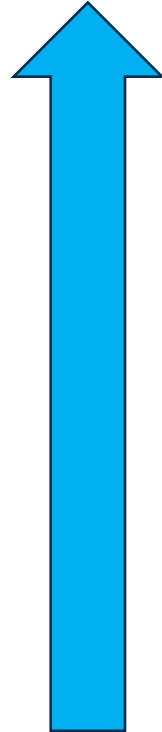
What counts Literature Review?

- Literature covers everything relevant that is written on a topic:
 - Journal articles,
 - Conference papers,
 - Text Books,
 - Theses and dissertations,
 - Newspaper articles,
 - Historical records,
 - Government reports, , etc.
- The important word is 'relevant'. Check with your supervisor or tutor when in doubt.

CHAPTER 2 – LITERATURE REVIEW

Good sources of Literature Review?

- Journal Papers
- Conference Papers
- Textbooks
- Other Books
- Company Whitepapers
- Company Websites
- Blogs
- Wikis



CHAPTER 2 – LITERATURE REVIEW

Research Papers

- A typical research paper (from a conference or journal) consists of the following parts;
 - Title,
 - Abstract,
 - Introduction,
 - Methodology,
 - Results and
 - Bibliography

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How many references to look?

- This actually depends on the level or stage at which the studies are.
- Normally, the following is recommended:
 - **Undergraduate review: 5-20 titles**
 - **Honor's dissertation: 20+ titles**
 - Masters thesis: 40+ titles
 - Doctoral thesis: 50+ titles

CHAPTER 2 – LITERATURE REVIEW

Writing a Review?

- Having gathered the relevant details about the literature, next is to write the review. The kind of review to write is as follows:
 - Research - to discover what has been written about the topic.
 - Critical Appraisal - to evaluate the literature, determine the relationship between the sources and ascertain what has been done already and what still needs to be done.
 - Writing - to explain what has been found.

CHAPTER 2 – LITERATURE REVIEW

Critical Appraisal (Examples)

- By critically reviewing the literature, the following items identified:
 - Some researchers have been worried only about particular characteristics of CPV and they have conducted their researches only on specific parameters and not all. Although it doesn't address the weaknesses of the research, but it can be concluded that for an optimal data acquisition system all the important parameters discussed so far must be acquired and monitored.
 - Display of data acquired on the site performed by either using LCD connected to the microcontroller or interfacing the DAQ board to the computer. The advantage of using LCD along with the microcontroller is that the readings can be done by anyone presented at the site.

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Critical Appraisal (Examples)

- By critically reviewing the literature, the following items identified:
 - Different methodology of database systems have been introduced for datasets storage(solar radiation, cell temperature, output current and output voltage), depends on the objective of the research project. For instance, for storing larger amount of variables/data, PC hard disk is preferred rather than EEPROM. However by directly storing data into the EEPROM there will be no requirement of dedicating one PC with proper software (such as LABVIEW) to do the same job and add up cost to the system.
 - Design and development of a microcontroller-based DAS system is influenced by the programming language such as assembly and C which is developed for the system.

CHAPTER 2 – LITERATURE REVIEW

Critical Appraisal (Examples)

- By critically reviewing the literature, the following items identified:
 - Development of DAQ system using imported boards (such as National instrument) is an imported technology for most developing countries and meaning a high cost and a barrier to the dissemination of such a system in CPV technology. Therefore, offering an alternative for commercial systems, which are more expensive, usually imported and have proprietary software is an outstanding proposal to solve this problem.
 - Mutually, all the researchers pointed at the effect of temperature on the efficiency and life-span of the semiconductor material used in CPV within their experimental result. Thus, the significance of this parameter urges to add an extra feature to the system to be able to indicate a signal for temperature rising above threshold.

CHAPTER 2 – LITERATURE REVIEW

Literature Review - Introduction

- The **introduction** should include:
 - The nature of the topic under discussion (the topic of your project).
 - The parameters of the topic (what does it include and exclude?).
 - The basis for the selection of the literature.

CHAPTER 2 – LITERATURE REVIEW

Literature Review - Body

- The **body** paragraphs could include relevant paragraphs on:
 - Findings by others.
 - Limitations and problems of the research by others,
 - Gaps in the topic,
 - Contribution to the body of knowledge that is relevant to the research problem,
 - How it compares and contrasts with the positions developed by other researchers,
 - Similar Systems,
 - Design/methodology developed by the researchers,

CHAPTER 2 – LITERATURE REVIEW

Literature Review - Body

- The **body** paragraphs could include relevant paragraphs on:
 - Current mainstream versus alternative theoretical or ideological viewpoints, including differing theoretical assumptions, differing political outlooks, and other conflicts, definitions in use,
 - Current research studies,
 - Current discoveries about the topic,
 - Principal questions that are being asked,
 - General conclusions that are being drawn,
 - Methodologies and methods in use.

CHAPTER 2 – LITERATURE REVIEW

Literature Review - Conclusion

- The **conclusion** should include:
 - A summary of major agreements and disagreements in the literature.
 - A summary of general conclusions that are being drawn.

CHAPTER 2 – LITERATURE REVIEW

Plagiarism

- To appropriate or pass off (the ideas or words of another) as one's own. [1]
- Growing problem with internet.
- An academic offence.
- Citing the reference at the end of a copied (or mostly copied) phrase/paragraph is still plagiarism.
- Use your own words.

[1] The New Penguin English Dictionary, Penguin, 1986

CRESWELL'S 5 STEPS TO CONDUCT A LITERATURE REVIEW

Step 1: Identify Key Terms or Descriptors [1]

- Extract key words from your title remember, you may decide to change the title later.
- Use some of the words other authors reported in the literature.

[1] Creswell, J.W. (2005) Educational Research: Planning, Conducting, and Evaluating Quantitative and Qualitative Research

CRESWELL'S 5 STEPS TO CONDUCT A LITERATURE REVIEW

Step 2: Locate Literature

- Use academic libraries, do not limit your search to an electronic search of articles.
- Use primary and secondary sources.
 - A primary source is research reported by the researcher that conducted the study. [1]
 - A secondary source is research that summarizes or reports findings that come from primary sources. [1]

[1] Creswell, J.W. (2005) Educational Research: Planning, Conducting, and Evaluating Quantitative and Qualitative Research

CRESWELL'S 5 STEPS TO CONDUCT A LITERATURE REVIEW

Step 2: Locate Literature

- It is best to report mostly primary sources.
- Search different types of literature:
 - Summaries, encyclopaedias, dictionaries and glossaries of terms, handbooks, statistical indexes, reviews and syntheses, books, journals, indexed publications, electronic sources, abstract series, and databases.

[1] Creswell, J.W. (2005) Educational Research: Planning, Conducting, and Evaluating Quantitative and Qualitative Research

CRESWELL'S 5 STEPS TO CONDUCT A LITERATURE REVIEW

Step 4: Organize the Literature

- Create a file or abstract system to keep track of what has been read. Each article that have been read should be summarized in one page containing.
 - Title (type the title so that it could be later copy-paste this into the References section of the report).
 - Source: journal article, book, glossary, etc.
 - Research problem: one or two lines will suffice Research Questions or Hypotheses.
 - Data collection procedure (a description of sample characteristics can be very handy as well).
 - Results or findings of the study.

CRESWELL'S 5 STEPS TO CONDUCT A LITERATURE REVIEW

Step 4: Organize the Literature

- Sort these abstracts into groups of related topics or areas which can then become the different sections of the review.

[1] Creswell, J.W. (2005) Educational Research: Planning, Conducting, and Evaluating Quantitative and Qualitative Research

CRESWELL'S 5 STEPS TO CONDUCT A LITERATURE REVIEW

Step 5: Write a Literature Review

- Types of reviews:
 - Thematic Review: A theme is identified and studies found under this theme are described. Major ideas and findings are reported rather than details.
 - Study-by-study Review: a detailed summary of each study under a broad theme is provided. Link summaries (or abstracts) using transitional sentences. Must be organized and flow coherently under various subheadings. Avoid string quotations (i.e. lengthy chunks of text directly quoted from a source).

[1] Creswell, J.W. (2005) Educational Research: Planning, Conducting, and Evaluating Quantitative and Qualitative Research

CAROL MOHRBACHER'S – LITERATURE REVIEW

Prewriting Stages

- Formulate problem or primary research question which topic or field is being examined and what are its component issues?
- Choose literature find materials relevant to the subject being explored and determine which literature makes a significant contribution to the understanding of the topic.
- Analyze and interpret the findings and conclusions of pertinent literature, how each contributes to the field opted.

CAROL MOHRBACHER'S – LITERATURE REVIEW

Organizing/Outlining

- Methods for organizing the Literature Review:
 - By subject (if it covers more than one subject)
 - Chronologically
 - By theme, idea, trend, theory, or major research studies
 - By author
 - By argumentative stance
- In all methods, relationships between elements (e.g., subject, theme, author, etc.) must be shown.

CAROL MOHRBACHER'S – LITERATURE REVIEW

Ongoing – housekeeping strategies and tips

- Immediately document and cite source from where the information is taken.
- Bookmark online sources.
- Bookmark - hard copy. Use post-it notes to mark pages with relevant information.
- Note any connections between sources in separate notebook or on post-its on pertinent pages.
- If information comes from a class lecture, interview, or conference, note details immediately.

CAROL MOHRBACHER'S – LITERATURE REVIEW

Beginning the Writing

- Introduce the Literature Review by:
 - Defining or explaining the primary problem addressed by the thesis and thus, by the sources chosen. OR
 - Explaining main conflict(s) in the literature. OR
 - Explaining the time frame in which review is done. OR
 - Offer a rationale for the choice of source material. OR
 - Using all or some of the points above.

CAROL MOHRBACHER'S – LITERATURE REVIEW

Body of the Literature Review

- Use subheadings if dividing the Literature Review topically, thematically, according to argumentative perspective, or according to time period.
- Be sure to show relationships between sources.
- Make explicit connections between reviewed sources and thesis.
- Discuss source's significant contributions.
- Do not develop ideas or use sources that are irrelevant to your thesis overall.
- References to prior studies should be in past tense; references to narrative or text other than studies should be in present tense.

CAROL MOHRBACHER'S – LITERATURE REVIEW

Concluding the Literature Review

- Summarize ideas, conflicts, themes, or historical (or chronological) periods.
- Contextualize the thesis topic within the summary.
- Point out gap(s) in scholarship and, show how the research helps to fill the gap(s).
- Transition to your next chapter.

CHAPTER 2 – LITERATURE REVIEW

- Topics covered, including the year, the industry, the country and/or region, and the following:
 - Findings by others
 - Limitations and problems of the research by others
 - Gaps in the topic
 - Contribution to the body of knowledge that is relevant to the research problem
 - How it compares and contrasts with the positions developed by other researchers
 - Similar Systems
 - Design/Methodology developed by the researchers

CHAPTER 2 – LITERATURE REVIEW

Styles and Samples of doing Literature Review

Title	
Intro	
Problem Statement	
Methodology	
Limitation	
Results	
Reference	
Citation	

CHAPTER 2 – LITERATURE REVIEW

Table 1 – Literature Review Summary

Year	Author	Method/technique	Microcontroller	Sensor	Communication device
1			8085	Temp	
2			8051	Temp	
3			PIC	Pressure	
10			Ard..	T, P & H	

CHAPTER 2 – LITERATURE REVIEW

Table 2 – Literature Review Summary (Results)

Year	Author	Accuracy	Recognition Rate	Power Saving	
1		70%			
2		82%			
3		85%			
10		93%			

CHAPTER 2 – LITERATURE REVIEW

Styles and Samples of doing Literature Review

Year	Author	Gripping Method	Advantage	Disadvantage

CHAPTER 2 – LITERATURE REVIEW

Table and Figure Formatting

Table 2.2: Comparison between RFID Technology and Tradition Way
(Minbo et al., 2011)

	Average time of receiving (minutes per note)	Average time of receiving (minutes per pallet)	Average time of picking (minutes per pallet)
Tradition technology	12.93	3.30	2.42
RFID	7.52	2.00	0.82
Efficiency Improvement	71%	65%	195%

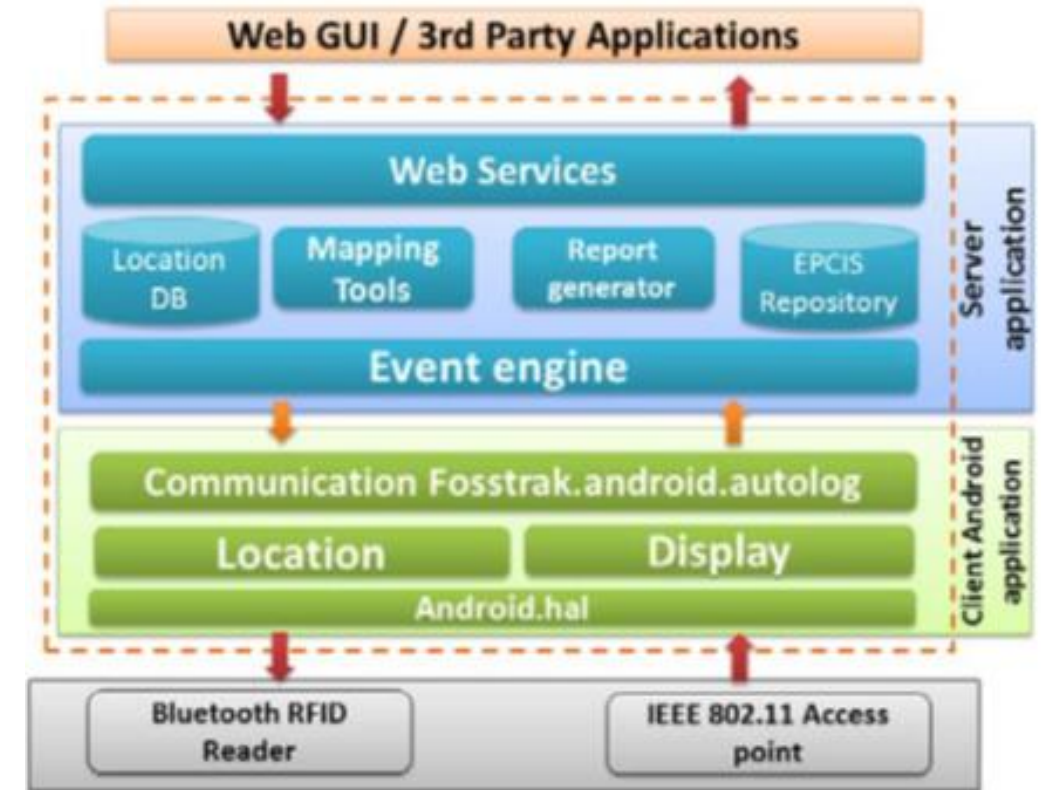


Figure 2.2: Modules and Communication diagram
(Pastor et al., 2011)

CHAPTER 2 – LITERATURE REVIEW-CRITICAL THINKING FRAMEWORK

Sample Project Title

Development of a Low-Cost Autonomous Mobile Robot with Enhanced Obstacle Detection for Dynamic Indoor Environments

Sample Research Problem Statement

Mobile robots used in indoor environments often experience unreliable navigation when exposed to dynamic, cluttered, or low-light conditions. Existing low-cost solutions frequently depend on basic sensors such as ultrasonic or infrared modules, which provide limited accuracy and slow response times when obstacles move unpredictably. High-precision solutions like LIDAR or stereo vision systems exist but are costly, computationally heavy, or unsuitable for undergraduate-level embedded platforms. This creates a research gap in designing a cost-effective yet robust navigation system that integrates multi-sensor fusion to improve obstacle detection, mapping, and path-planning performance in real time. Therefore, this project investigates the limitations of current indoor mobile robot navigation methods and explores improved design approaches through systematic literature review and a 5W1H critical thinking framework.

CHAPTER 2 – LITERATURE REVIEW-CRITICAL THINKING FRAMEWORK-5W1H

W1H Element	Guiding Question	Critical Insight (Based on Literature Review)
WHAT	What is the problem?	Mobile robots face unreliable obstacle detection and navigation in dynamic indoor environments, leading to collisions and poor path planning.
WHY	Why is it a problem?	Indoor environments continuously change—people move, objects shift, and lighting varies—making low-cost sensors insufficient for safe real-time navigation.
WHO	Who is affected?	Robotics students, researchers, service-robot users, and institutions deploying affordable automation solutions.
WHERE	Where does it occur?	Indoors: classrooms, labs, corridors, apartments, hospitals, and community centres.
WHEN	When does it happen?	During real-time robot operation, especially under low-light, cluttered layouts, or sudden obstacle movement.
HOW	How is it currently addressed?	Through SLAM, vision-based mapping, ultrasonic/IR fusion, and deep learning—but each has limitations in cost, accuracy, or computation.

CHAPTER 2 – LITERATURE REVIEW

- Summarize the entire chapter.

QUICK REVIEW QUESTIONS

- What is Literature Review?
- Why do we need literature review?
- What counts literature?

SUMMARY OF MAIN TEACHING POINTS

- Introduction
- Literature Review

QUESTION & ANSWER SESSION

Q & A

WHAT WE WILL COVER NEXT

- Proposed Methodology